GPR 4-Arm Spiral Antenna Project Notes

Prepared summary for student discussion and design review

This document summarizes the previous CST antenna discussion and simulation work. The antenna under study is a spiral antenna intended for Ground Penetrating Radar (GPR) operation around 433 MHz and 866 MHz. The current result is not final; it is an intermediate design that still needs optimization because extra unwanted resonance dips/peaks are appearing in the S-parameter response.

1. Project Objective

The goal of this project is to design and simulate a compact spiral antenna structure for GPR applications. The main operating frequencies considered are 433 MHz and 866 MHz. The design is being built and checked in CST Microwave Studio before fabrication.

- Target application: Ground Penetrating Radar (GPR).
- Target bands: approximately 433 MHz and 866 MHz.
- Main antenna type: spiral / log-spiral antenna.
- Simulation target: good impedance matching near the required frequencies and reduced unnecessary dips between them.
- Final practical target: a design that can be fabricated and later fed using a coaxial/balanced feed arrangement.

2. Why GPR Needs This Type of Antenna

In GPR, electromagnetic waves are transmitted into the ground and reflected from buried objects, voids, soil layers, or material discontinuities. The antenna should therefore support useful penetration and sufficient bandwidth while keeping ringing and unwanted coupling low.

- Lower frequency such as 433 MHz gives better ground penetration but requires larger antenna size.
- Higher frequency such as 866 MHz gives better resolution but lower penetration depth.
- Using both frequency regions helps balance penetration and resolution.
- A smooth time-domain response is important in GPR because unnecessary ringing can hide the actual reflected signal.

3. Why Spiral Antenna Was Chosen

A spiral antenna was selected because it can support wideband or multi-band behavior and can be made as a planar structure, which is suitable for CST modeling and possible PCB-style fabrication.

Antenna type	Advantages	Reason not selected / limitation
Spiral antenna	Planar, wideband, can support circular polarization, suitable for 433–866 MHz study	Selected for this project
Vivaldi antenna	Very wideband and directional	Can become large at 433 MHz and may need careful taper/feed design
Horn antenna	High gain and stable radiation	Bulky and not convenient for compact GPR setup

Bow-tie antenna	Simple and wideband	May need additional backing/absorber and can have ringing or pattern issues

For this work, the spiral antenna is preferred because it is planar, easier to modify parametrically, and suitable for studying dual-frequency or wideband GPR behavior.

4. Basic Design Idea

The initial antenna design was a 4-arm spiral/log-spiral structure. The arms are placed around the center with phase-related feeding. The main design variables are inner radius, outer radius, number of turns, arm width, spacing, substrate size, absorber position, and feed-port arrangement.

5. Basic Formula References

Only upper-level formulas are shown here. Detailed derivation is not included because the purpose is project explanation, not full mathematical proof.

- Free-space wavelength: lambda = c / f
- Effective wavelength: lambda_eff = lambda / sqrt(eps_eff)
- Approximate active spiral radius: r_active = lambda_eff / (2*pi)
- Log spiral form: r = r_in * exp(b * theta)
- Log spiral growth factor: b = ln(r_out/r_in) / theta_max
- Total angular length: theta_max = number_of_turns * 2*pi

For the earlier design, the effective permittivity was approximated around eps_eff = 1.10. FR-4 relative permittivity was considered around 4.4, but the effective value is lower because the field is partly in air and partly near the substrate/ground environment.

6. Material Selection

Material	Typical value used	Purpose
Copper / Copper annealed	Conductivity approx. 5.8e7 S/m, thickness 0.035 mm	Spiral conducting arms
FR-4 lossy substrate	epsilon_r approx. 4.4, tan delta approx. 0.02, thickness 1.6 mm	Mechanical support and PCB fabrication base
Lossy absorber	epsilon approx. 8–10, tan delta approx. 0.15–0.5	Reduce ringing and unwanted reflections
Soil block	Depends on soil moisture; example epsilon around 9 and conductivity around 0.01 S/m	GPR environment simulation
Air / open boundary	Open/PML boundary	Represents free-space around antenna

7. Substrate, Soil, and Absorber Setup

The antenna was placed on an FR-4 substrate. A lossy absorber was tested behind the antenna to suppress ringing and reduce unwanted resonances. Soil was also considered below/behind the antenna to represent the GPR environment.

- Substrate: circular FR-4 disk or board, thickness about 1.6 mm.
- Absorber: placed behind the antenna, typically 5–15 mm thick in tests.

- Absorber gap: a small air gap was used between substrate and absorber, for example 2–5 mm.
- Soil block: added below absorber or antenna region when simulating GPR ground interaction.
- Important rule: absorber and soil should not unintentionally overlap unless that is intended in the model.

8. Port and Feeding Concept

Several feeding approaches were discussed. The initial model used multiple ports for the 4-arm structure. This can create multiple S-parameter responses and may contribute to extra dips if the excitations are not combined properly.

- A 4-arm spiral can use four individual ports, but the interpretation of combined S11 becomes more complex.
- A practical 4-arm design usually requires a controlled phase feed, such as 0°, 90°, 180°, and 270° excitation.
- For two opposite pairs, two differential ports can be used: one port between one pair of opposite arms and another port between the other pair.
- For circular polarization, a quadrature feed is required. This normally means equal amplitude and 90° phase difference.
- In fabricated hardware, a coaxial feed alone may need a balun, hybrid coupler, or feed network to create the required balanced/phase-shifted excitation.

For later fabrication, a coaxial cable can be used as the input connection, but the feed must be designed carefully. If the antenna is differential, the coaxial feed should be converted properly using a balun or feed network. If a 4-arm circularly polarized feed is used, a 90-degree hybrid or quadrature network is typically needed.

9. Current Simulation Result and Observation

The attached CST result shows the earlier antenna response. The design gives a good dip near 433 MHz and a moderate dip near 866 MHz, but it also contains several additional unwanted dips between the two target frequencies. These unwanted dips indicate extra resonances and possible ringing paths in the antenna/cavity/absorber/feed arrangement.

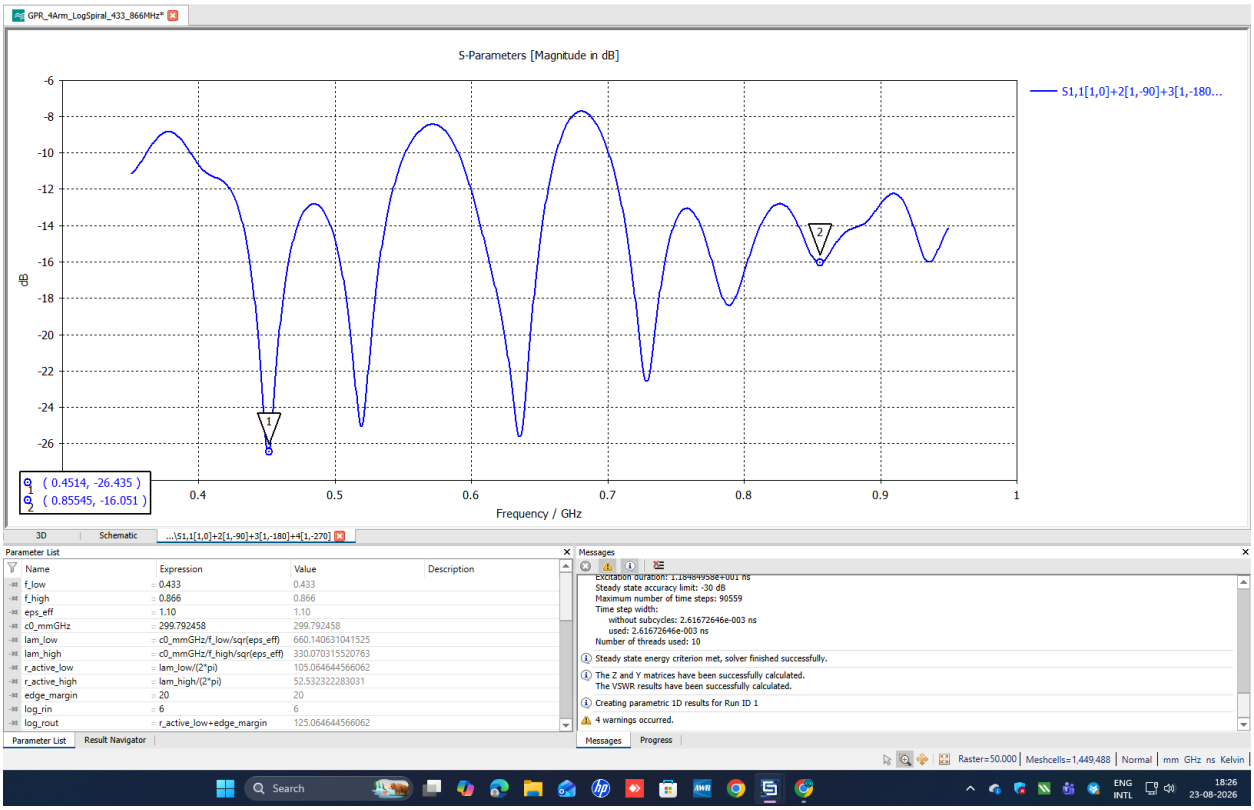


Figure: Previous CST model and S-parameter result showing desired and unwanted resonances.

- One useful dip was observed near 433 MHz.
- A second response appeared near 866 MHz, but it was not always deep enough.
- Multiple unwanted dips appeared in the middle frequency range.
- These extra dips are not the final desired result and need further optimization.

10. Why Extra Peaks/Dips May Appear

- Multiple arms and ports can create several current paths and resonant modes.
- The spiral structure is naturally wideband/multiresonant, so extra dips can appear.
- Cavity depth, absorber position, and air gap can introduce additional standing-wave effects.
- FR-4 loss, soil loading, and absorber loss can shift the resonant frequencies.
- Feed impedance and phase combination strongly affect the final S-parameter curve.

11. Optimization Direction

The current result is not final. The next work is to optimize the antenna response so that useful behavior appears near 433 MHz and 866 MHz while unnecessary middle dips are suppressed or reduced.

1. Tune spiral outer radius and number of turns to control low-frequency behavior near 433 MHz.
2. Tune inner radius, feed pad, and port impedance to control high-frequency behavior near 866 MHz.
3. Adjust absorber loss and absorber spacing to reduce ringing without killing useful resonance.
4. Compare 4-port, 2-port differential, and final coaxial/balun feed approaches.
5. Check S11, S22, coupling terms, radiation pattern, and time-domain ringing, not only one S11 curve.
6. After simulation optimization, finalize a fabrication-safe layout with minimum copper gap and practical feed connection.

12. Fabrication Notes

- Copper width and spacing must follow PCB fabrication limits; keep clear gaps between arms.
- Avoid overlapping copper arms or common center shorting unless intentionally designed.
- If coaxial feed is used, the transition from coax to spiral must be mechanically and electrically stable.
- For balanced spiral feeding, a balun or hybrid coupler should be considered.
- Before fabrication, verify the final design with CST mesh, port setup, boundary conditions, and material stackup.

13. Conclusion

The project is focused on developing a spiral antenna for GPR operation around 433 MHz and 866 MHz. Spiral geometry was selected because it is planar, wideband, and suitable for circular-polarization-oriented feed design. The previous CST result shows promising resonance behavior but also contains unwanted dips. Therefore, the current design should be treated as an intermediate prototype. Further work is needed to optimize arm geometry, absorber placement, port/feed network, and fabrication layout before final hardware implementation.

Important note: The current CST result is not final. Extra peaks/dips must be reduced by optimization before fabrication.